Nazmul Haque

Gazipur, Bangladesh
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RESEARCH INTEREST

Multivariate Analysis, Copula Modeling, Bioinformatics, Biostatistics, Environmental Data Science, Machine Learning

EDUCATION

Shahjalal University of Science and Technology, Sylhet, Bangladesh

***MS** in Statistics* | 2022-2023

> **MS Thesis:** Modeling Correlated Flood Peak and Volume with Copula-Based Joint Distribution: Improving Flood Risk Estimation in Bangladesh; Supervisor: Dr. Khalidur Rahman

***BSc** in Statistics* | 2018-2021

PROFESSIONAL EXPERIENCE

Bangladesh Rice Research Institute, Gazipur, Bangladesh

Scientific Officer, Agricultural Statistics Division | 3 Jun 2026 – Present

- Conduct statistical analysis, data management, and research support for agricultural and climate-related studies.
- Apply statistical and machine learning methods to analyze experimental, spatial, and weather-related agricultural data.

RESEARCH EXPERIENCE

Bangladesh Rice Research Institute, Gazipur, Bangladesh

Research Assistant | Aug 2025-31 May 2026

- Apply machine learning and deep learning algorithms (MLP, CNN, LSTM) for weather forecasting.
- Perform spatial analysis (IDW, IK, OK, Moran's I) to detect rice neck blast hotspots

Department of Statistics, SUST, Sylhet, Bangladesh

Research Assistant, PI: Dr. Khalidur Rahman | July 2024-Jun 2025

- Estimated the marginal distributions of flood phenomena such as flood peak, volume, rainfall
- Developed copula-based joint distributions capturing tail dependence
- Calculated return period of flood using bivariate and tri-variate copula function.

Research Assistant, PI: Dr. Md. Azizul Baten | July 2023-Jun 2024

- Designed questionnaire for primary data collection
- Analyzed climate pattern for haor region (northeast part of Bangladesh)
- Predicted vulnerability prediction using Machine learning algorithms (logistic, DT, RF, SVM, KNN)

TEACHING AND TRAINING EXPERIENCE

Sher-e-Bangla Agricultural University, Dhaka, Bangladesh

Trainer, R Programming for Research Data Analysis | 2-Day Training Program for MS and PhD Students | 8–9 May 2026

- Conducted hands-on R programming training focused on experimental design and multivariate data analysis.
- Delivered practical sessions on Completely Randomized Design, Randomized Complete Block Design, and Principal Component Analysis using R.

Sher-e-Bangla Agricultural University, Dhaka, Bangladesh

Assistant Trainer, R Programming for Research Data Analysis | 3-Day Training Program for Teachers | 27–29 November 2025

- Assisted in conducting hands-on R programming training for university teachers.
- Supported participants during practical sessions on data analysis, R coding, and troubleshooting.

PUBLICATIONS

(1) Rahman, N.M.F., **Haque, N.,** Akhtar, P. *et al.* Regional climate variability and rice yields prediction in Bangladesh using machine learning and deep learning models. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-73994-7

(2) Rashid, M. M., Rahman, N. M. F., **Haque, N.,** Khan, M. A. I., Mian, M. S., Hossain, M., ... & Islam, M. R. (2026). Delineating season-specific hotspots: A spatio-temporal risk assessment of neck blast disease in rainfed and irrigated ecosystems of Cumilla District, Bangladesh. PLoS One, 21(9), e0358537.

(3) Rahman, N.M.F., **Haque, N.,** Asadullah, M. *et al.* Machine learning-based temperature prediction across diverse ecosystems for the Boro Season in Bangladesh. *Sci Rep* **16**, 15437 (2026).

(4) **Haque, N.,** Rahman, K., Hossain, M. S., & Alam, S. (2026). Copula-based joint distribution of flood peak and volume for improved flood risk assessment: a case study of the Surma-Meghna River Basin, Bangladesh. *Journal of Water and Climate Change*, 17(1), 1-18.

(5) **Haque, N.,** Haque, M. A., Sadikeen, S. S., & Uddin, M. J. (2025). A hierarchical competing risks survival model in the study of child mortality in Bangladesh. *Scientific Reports*, *15*(1), 44812.

(6) Rahman, N. M. F., Asadullah, M., Hossain, M. S., Uddin, M. J., **Haque, N.,** Rashid, M. B., ... & Baten, M. A. (2025). Spatiotemporal trends and machine learning-based prediction of temperature variability during the T. Aman rice-growing season in Bangladesh. *Scientific Reports*, *15*(1), 44883.

(7) Sadikeen, S. S., **Haque, N.,** Hossain, M. M., & Uddin, M. J. (2024). Impact of food price inflation on stunting in under five aged children in Bangladesh. *Health Economics Review*, *14*(1), 6. PMCID: PMC11363589

PUBLICATIONS (PREPRINT AVAILABLE)

(1) **Haque, N.,** Mazed, A., Ankhi, J. N., & Uddin, M. J. (2026). TF-IDF k-mer–based Classical and Hybrid Machine Learning Models for SARS-CoV-2 Variant Classification under Imbalanced Genomic Data*. bioRxiv*, 2026-04. https://doi.org/10.64898/2026.04.02.716024

(2) Alam,S., Haque,N., Mazed,A., & Ahmed,F. (2026). Exchangeable Versus Age- structured Priors in Small-Area Estimation of Adult Mortality From Sibling Survival Histories. *SocArXiv*. DOI: 10.31235/osf.io/9dh8f_v1.

(3) Alam,S., Haque,N., Mazed,A., Mayen, M.S.A., Kowser, M.A., & Hassan, M. (2026). Network-Adjusted Empirical Bayes Estimation of Age-Specific Adult Mortality from DHS Sibling Survival Histories: Evidence from Nigeria. *Research Square*. DOI: 10.21203/rs.3.rs-10647369/v1

PUBLICATIONS (SUBMITTED)

(1) **Haque, N.,** Talukdar, U.S, & Baten, M. A., (2025). Assessing Climate Vulnerability of Haor Wetlands in Bangladesh Using Stakeholder Perceptions, GIS, and Machine Learning. *Submitted to GeoJournal*.

(2) **Haque, N.,** Rahman, K., (2025). Advancing Flood Frequency Analysis through Tri-variate Copulas: The Surma–Meghna River Basin Case Study. *Submitted to Scientific Reports*.

SOFTWARE SKILLS

R Studio, Python, SPSS, SAS, MySQL, STATA, Excel

HONORS AND AWARDS

- National Science and Technology (NST) Fellowship 2022-23, Ministry of Science and Technology, Government of the People's Republic of Bangladesh.
- SUST Merit Grant for Academic Results
- Board Scholarship, Junior School Certificate Examination

TRAINING AND WORKSHOPS

- **8th International Conference on ICRSDS4IR**, Rajshahi University — Presented research (Dec 2024)
- **Workshop on Research Methodology and Scientific Publication**, SUST, Sylhet (Sept 2023)
- **Industrial Field Tour**, Paramount Textile PLC, Gazipur— Statistical tools and methods in textile industries (Sept 2022)
- **Field Work on Agricultural Experimental Design**, BRRI, Gazipur (Oct 2019)

EXTRACURRICULAR ACTIVITIES

- Joint Secretary, Sirajganj Association of SUST (2021-2023)
- Volunteer, Youth Innovation Leadership and Sustainable Development (2021-2022)
- Volunteer and Presented several Poster, International seminar and StatFest at SUST(2023)